

Cybersecurity & Ethical Hacking

Task-2 Report

Objective:

The objective of this task is to help you build a safe and isolated cybersecurity practice environment on your own computer. This hands-on lab will serve as the foundation for all future tasks such as vulnerability assessment, web application testing, exploitation practice, and incident response.

I will create:

- 1 Attacker machine (Kali Linux)
- 1 Vulnerable target application/machine (DVWA or OWASP Juice Shop)
- Proper network segmentation (NAT + Host-Only)

What I Have Learn

Skill Area	Practical Outcome
Virtualization	Running multiple isolated systems for pentesting
Lab isolation	Preventing real-world system damage
Web targets setup	Deploying intentionally vulnerable applications
Basic networking	NAT vs Host-Only and subnet usage
Tool awareness	Nmap, Burp, Wireshark validation

Prerequisites

Requirement	Recommendation
RAM	Minimum 8GB, Preferably 16GB
Disk Space	40-60 GB free
CPU	Virtualization enabled
OS	Windows / Mac / Linux

Procedure

Step 1: Install VirtualBox / VMware

- Create Host-Only Network and enable NAT.

Step 2: Deploy Kali Linux (Attacker VM)

- Import VM image, assign 2 CPUs & 3–4GB RAM.
- Configure two adapters: NAT + Host-Only.
- Update system:
`sudo apt update && sudo apt upgrade -y`
- Create working folders:
`mkdir -p ~/lab/{scans,notes,pcaps,screenshots}`

Step 3: Deploy Target Machine

- **Option A (Juice Shop via Docker):**
`sudo apt install -y docker.io`
`sudo systemctl enable --now docker`
`sudo docker pull bkimminich/juice-shop`
`sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop`
- **Option B (DVWA):** Setup Apache, PHP, MariaDB, clone DVWA repo.

Step 4: Verify Connectivity

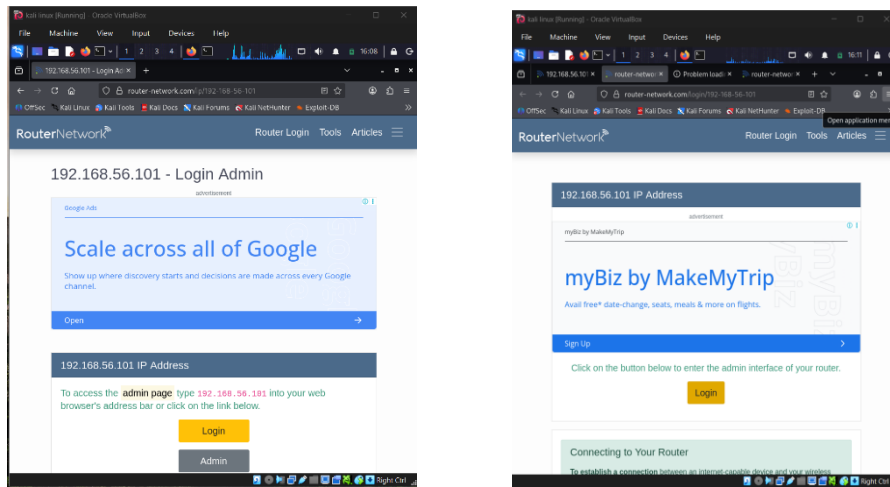
- Run ping sweep:
`nmap -sn 192.168.56.0/24`
- **Confirm attacker detects target.**

Step 5: Validation

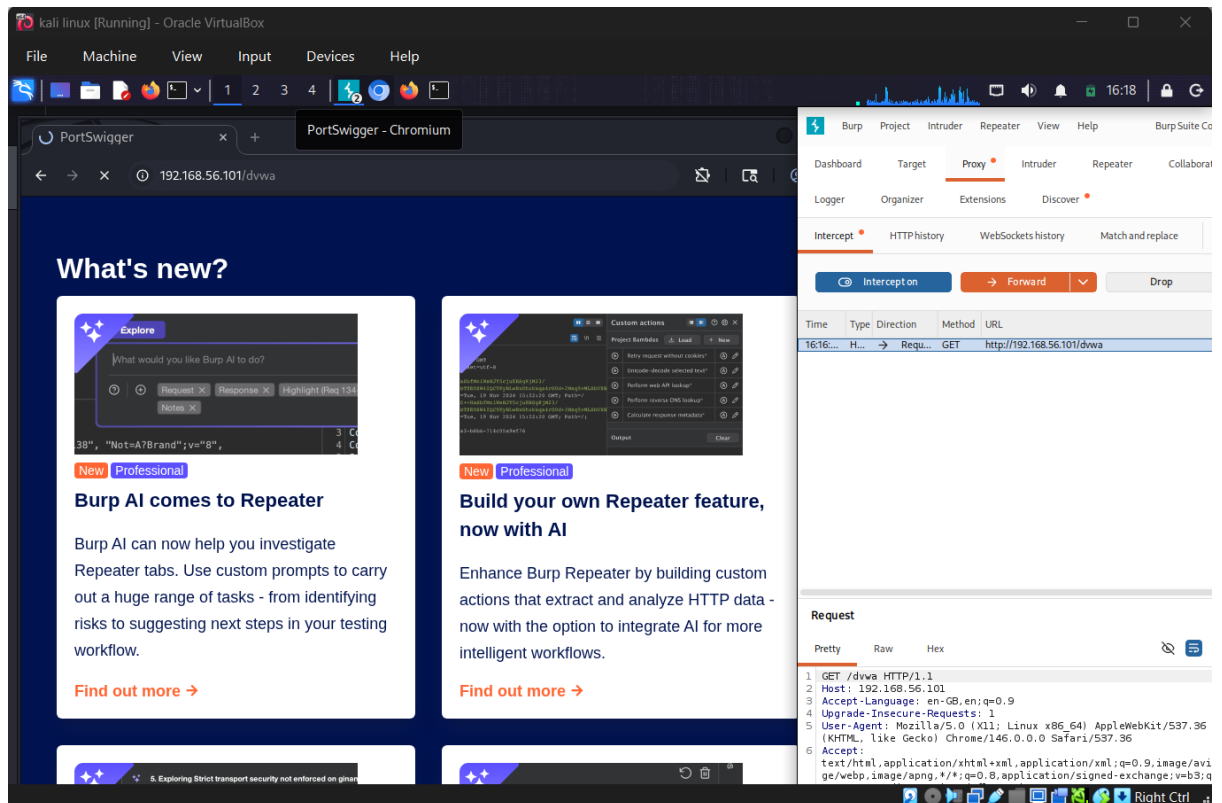
- **Access target web app in browser.**
- **Intercept traffic with Burp Suite.**
- **Capture packets with Wireshark.**
- **Save screenshots in ~/lab/screenshots.**

Basic Validation (Proof)

1. Access target web app in browser



2. Open Burp and intercept traffic

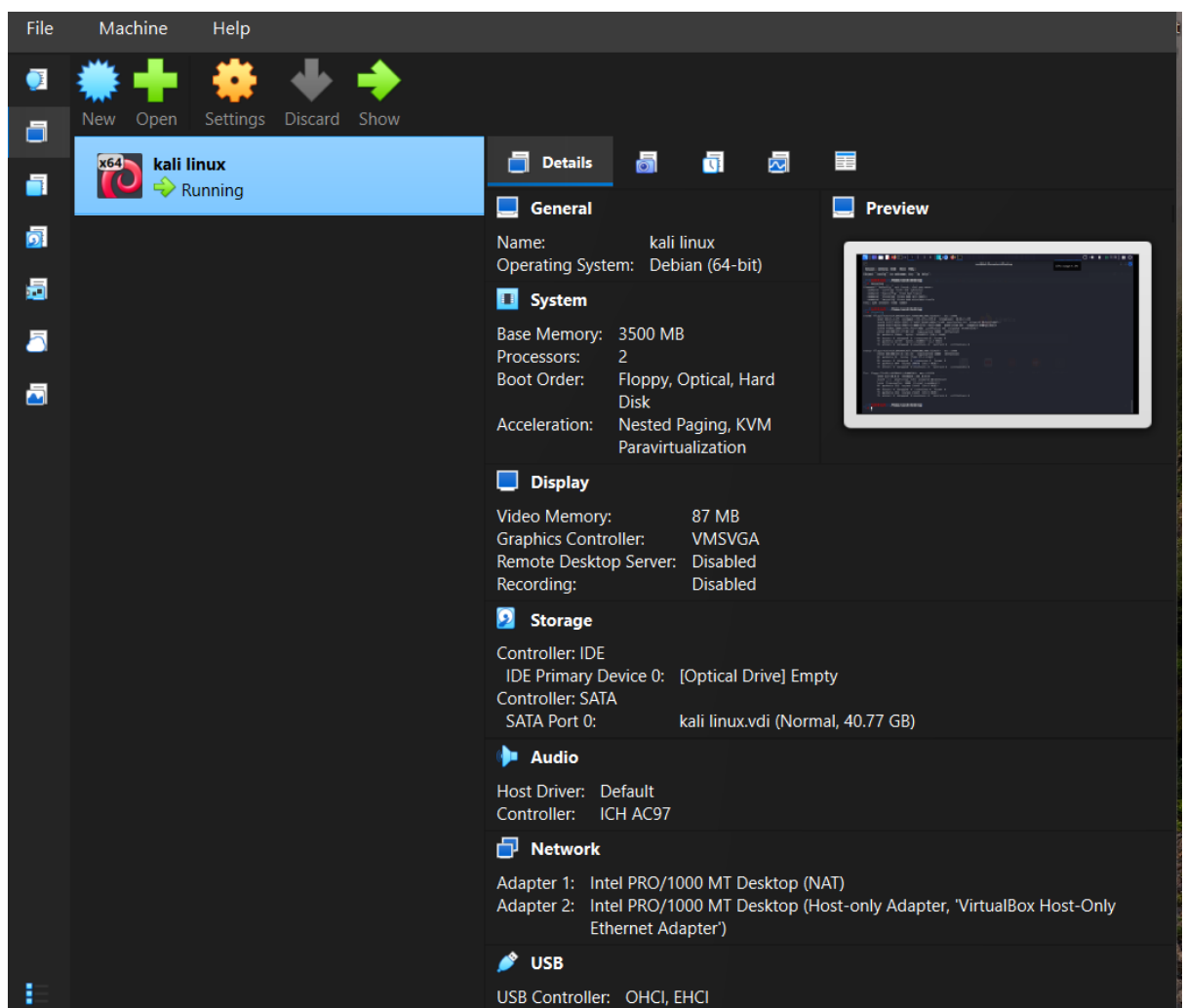


3. Run a simple Nmap Scan

```
(root@kali)-[/home/vansh/Desktop]
# nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 15:59 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00048s latency).
MAC Address: 0A:00:27:00:00:08 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00056s latency).
MAC Address: 08:00:27:CC:2F:EB (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 1.91 seconds
```

Observations (Evidence Placeholders)

- Figure 1: VM List



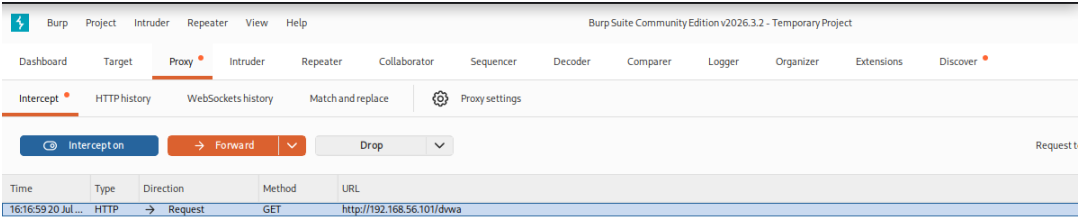
• **Figure 2: IP Configuration**

```
(root@kali)-[/home/vansh/Desktop]
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:27:09:dd brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82561sec preferred_lft 82561sec
    inet6 fd17:625c:f037:2:1d62:8468:ad5a:1ca8/64 scope global temporary dynamic
        valid_lft 86164sec preferred_lft 14164sec
    inet6 fd17:625c:f037:2:a00:27ff:fe27:9dd/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86164sec preferred_lft 14164sec
    inet6 fe80::a00:27ff:fe27:9dd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:dc:61:46 brd ff:ff:ff:ff:ff:ff
```

• **Figure 3: Nmap Scan Results**

```
(root@kali)-[/home/vansh/Desktop]
# nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 15:59 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00048s latency).
MAC Address: 0A:00:27:00:00:08 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00056s latency).
MAC Address: 08:00:27:CC:2F:EB (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 1.91 seconds
```

• **Figure 4: Burp Suite Interception**



```
Request
Pretty Raw Hex
1 GET /dvwa HTTP/1.1
2 Host: 192.168.56.101
3 Accept-Language: en-GB,en;q=0.9
4 Upgrade-Insecure-Requests: 1
5 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
6 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
7 Accept-Encoding: gzip, deflate, br
8 Connection: keep-alive
9
10
```

Conclusion

The successful completion of Task-2 marks the establishment of a fully functional and isolated cybersecurity lab environment, which is a critical foundation for ethical hacking practice. By deploying an attacker machine (Kali Linux) alongside a deliberately vulnerable target application (DVWA or OWASP Juice Shop), the lab provides a controlled ecosystem where penetration testing techniques can be explored without risking real-world systems. The integration of NAT and Host-Only networking ensured both internet connectivity for updates and strict traffic isolation for safe experimentation, thereby reinforcing the importance of network segmentation in cybersecurity.

Throughout the process, several essential skills were developed. The installation and configuration of virtualization software demonstrated the ability to manage multiple operating systems simultaneously, while the deployment of vulnerable applications highlighted the role of intentionally insecure environments in training. The use of tools such as Nmap for host discovery, Burp Suite for web traffic interception, and Wireshark for packet analysis provided practical exposure to industry-standard utilities, bridging theoretical knowledge with hands-on application. Each validation step — from confirming connectivity to capturing traffic — reinforced the iterative nature of cybersecurity testing, where observation and verification are as important as exploitation.

This lab also emphasized professional practices such as documentation, evidence collection, and structured reporting. Screenshots of VM lists, IP configurations, Nmap scans, Burp Suite interceptions, and Wireshark captures serve not only as proof of execution but also as a valuable reference for future tasks. The systematic organization of these deliverables mirrors real-world penetration testing engagements, where clear records are essential for accountability, reproducibility, and communication with stakeholders.